Highlights

- Imputation, feature selection and stacking were all fitted inside validation.
- Stacking did not beat logistic regression (CV AUC 0.843 vs 0.842).
- RFECV cut inputs by 25%, with a stable feature core of glucose and BMI terms.
- Probabilities were too high, consistent with class weighting.
- Findings replicated in an independent NHANES cohort with an HbA1c outcome.